

# Movie & Streaming Content Analytics

## Project Report

### 1. Project Overview

This project analyzes a movie and streaming-content dataset to understand content performance, ratings, popularity, audience engagement, genre patterns, and changes in the content catalog over time. The analysis is presented through a Power BI dashboard with year and genre filters.

### 2. Dataset

The supplied Excel dataset contains **9,837 records** and **9 columns**. The dataset includes title-level information used for the dashboard, including release date, title, overview, popularity, vote count, vote average, original language, genre, and poster URL.

**Fields:** Release\_Date, Title, Overview, Popularity, Vote\_Count, Vote\_Average, Original\_Language, Genre, Poster\_Url

### 3. Analysis Approach

The project follows a simple analytics workflow:

- Load and inspect the movie-content dataset.
- Prepare the fields required for analysis and visualization.
- Analyze ratings, popularity, votes, genres, languages, and release years.
- Build calculated measures and visual comparisons for the dashboard.
- Present the results through an interactive Power BI dashboard.

### 4. Dashboard Overview

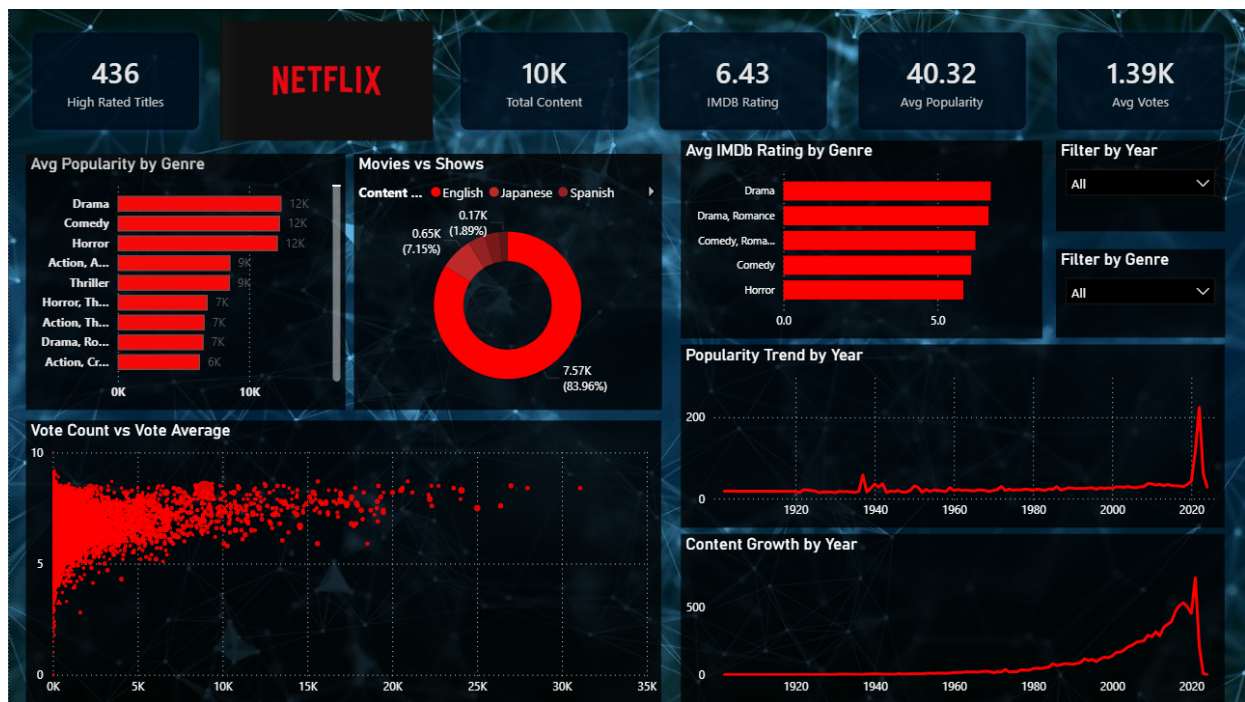


Figure 1. Movie & Streaming Content Analytics Dashboard

### 5. Dashboard KPIs

| Metric              | Dashboard Value |
|---------------------|-----------------|
| High Rated Titles   | 436             |
| Total Content       | 10K             |
| Average IMDB Rating | 6.43            |
| Average Popularity  | 40.32           |
| Average Votes       | 1.39K           |

### 6. Dashboard Analysis

**Average Popularity by Genre:** Compares popularity across genres to identify genres with stronger average popularity within the dataset.

**IMDb Rating by Genre:** Compares average ratings between genres and helps identify differences in content rating performance.

**Movies vs Shows / Language Distribution:** Shows the content mix and the distribution of original languages represented in the dashboard.

**Vote Count vs Vote Average:** Compares audience voting activity with average ratings to provide another view of content engagement.

**Popularity Trend by Year:** Tracks changes in average popularity across release years.

**Content Growth by Year:** Shows how the amount of content changes across the release timeline.

## 7. Key Business Questions

- Which genres have higher average popularity?
- Which genres have stronger average IMDb ratings?
- Which titles fall into the dashboard's high-rated group?
- How does audience voting relate to average ratings?
- How has content volume changed over time?
- How has popularity changed across release years?
- How does the content mix vary by language and genre?

## 8. Dashboard Filters & Interaction

The dashboard includes filters for **Year** and **Genre**. These allow users to narrow the analysis and compare different parts of the catalog without working directly with the raw dataset.

## 9. Tools & Technologies

**Excel** — source dataset   **Python** — data preparation and exploratory analysis   **SQL** — structured analysis   **Power BI** — dashboard and visualization

## 10. Project Outcome

The project converts a large movie-content dataset into an interactive analytical dashboard. It brings ratings, popularity, votes, genre performance, language distribution, and time trends together in one place, making the dataset easier to explore and interpret.

## 11. Project Deliverables

| File                       | Purpose                                                          |
|----------------------------|------------------------------------------------------------------|
| Business Problem Statement | Defines the project scope, objectives, and analytical questions. |
| Python Notebook            | Data preparation and exploratory analysis.                       |
| SQL Queries                | Structured data analysis.                                        |
| Power BI Dashboard         | Interactive visualization and filtering.                         |
| Project Report             | Documents the dataset, methodology, dashboard, and findings.     |
| README.md                  | GitHub project documentation.                                    |